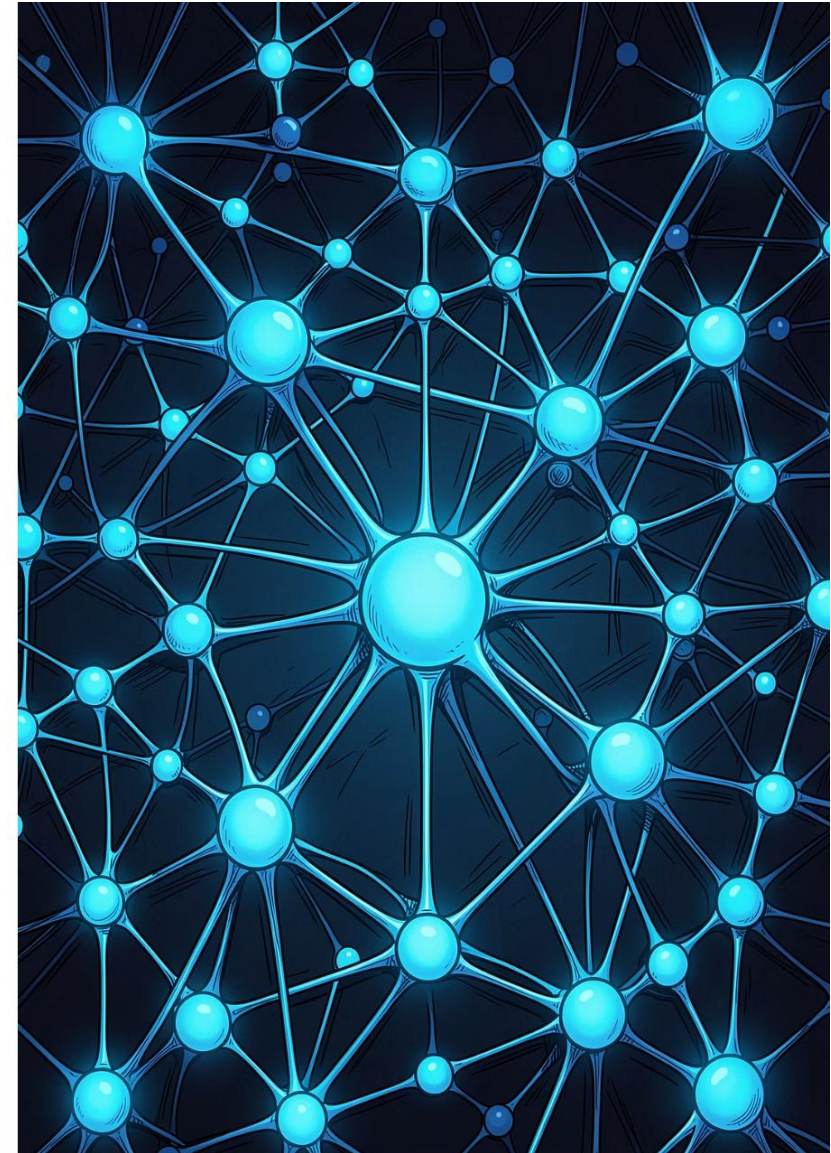


P15 - CARTOGRAPHERS OF THE INVISIBLE

Semantic Space Mapping using Sentence-BERT Embeddings

Natural Language Processing

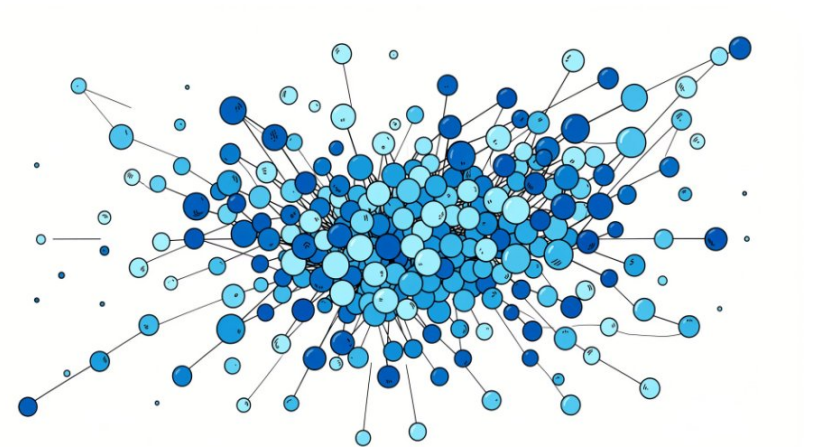
Karan Singh



Introduction

Semantic space mapping represents textual concepts as **numerical vectors** in a high-dimensional space — where semantically similar concepts are positioned **closer together**.

📄 **Research Question:** Can pretrained Sentence-BERT embeddings automatically organize textual concepts into meaningful semantic groups?

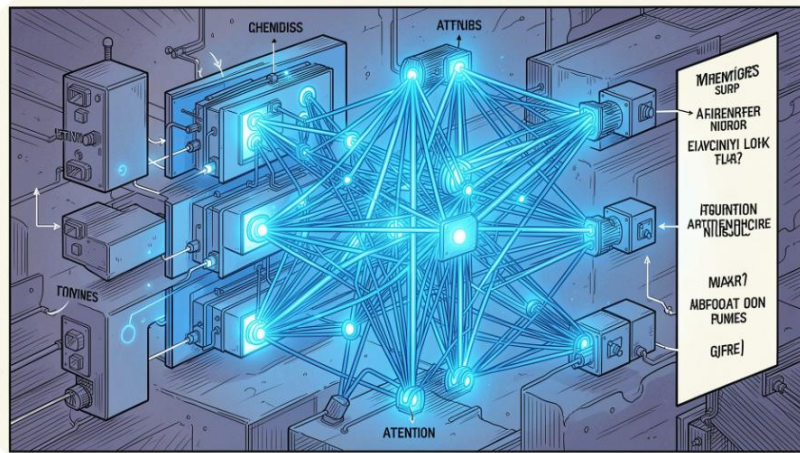


Dataset

The dataset comprises **50 concepts** spanning **5 semantic categories**, each paired with a short textual description used as input to the embedding model.

Sports	Food	Profession	Emotion	Technology
Football	Pizza	Doctor	Joy	Smartphone
Swimming	Sushi	Engineer	Sadness	Laptop
Tennis	Pasta	Teacher	Anger	Robot

Embedding Extraction with Sentence-BERT



The project uses **all-MiniLM-L6-v2**, a pretrained Sentence-BERT model that generates **contextual semantic embeddings** – moving beyond word frequency to capture meaning.



Contextual

Captures meaning in context



Dense Vectors

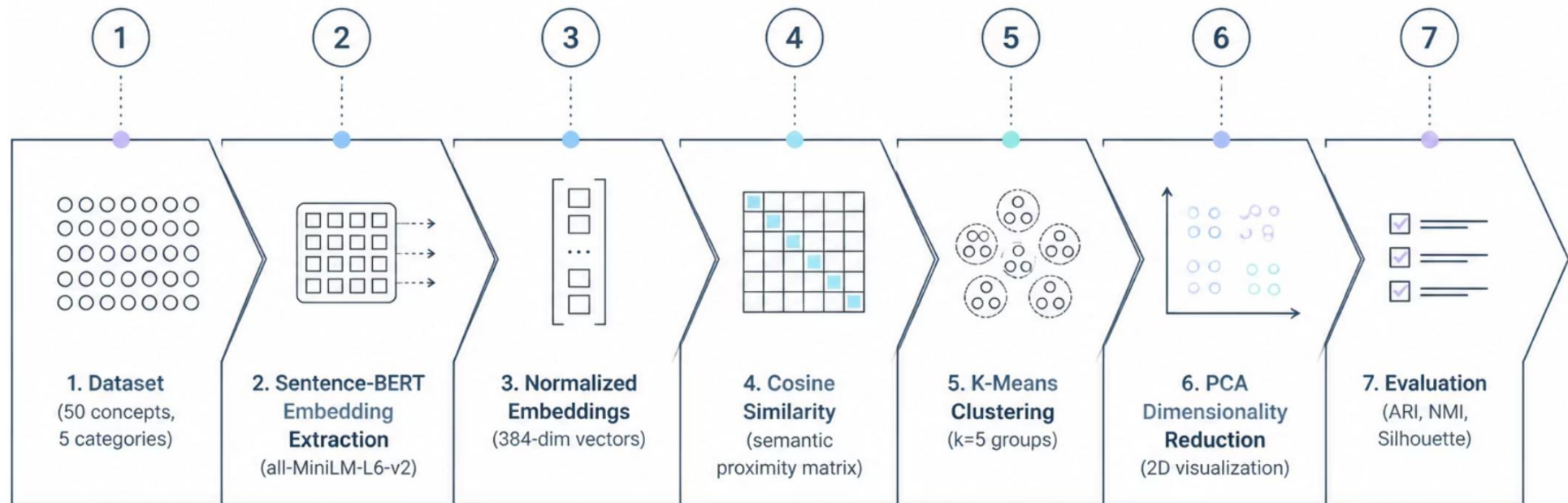
384-dimensional normalized output



Semantic

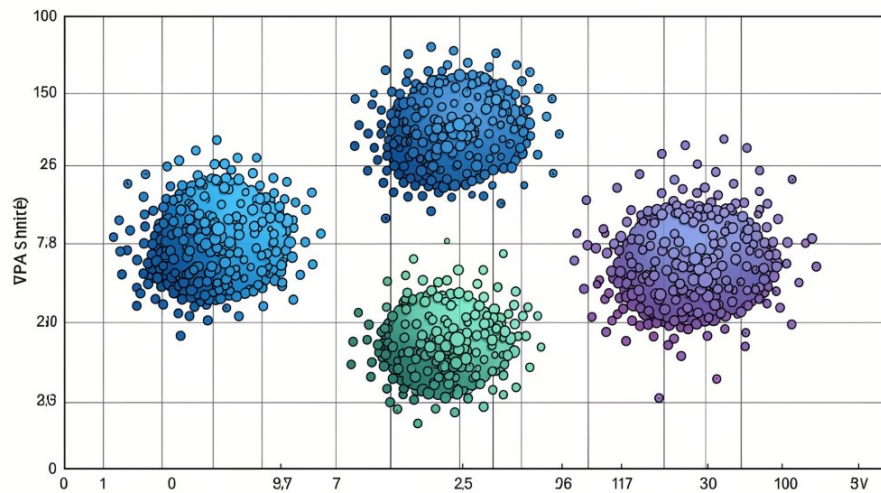
Optimized for similarity tasks

Methodology



Each stage transforms the data progressively – from raw text to interpretable 2D semantic clusters evaluated against ground-truth labels.

Similarity, Clustering & Visualization



Cosine Similarity

Measures semantic proximity between embeddings.
Values near **1.0** indicate high similarity.

K-Means Clustering

Groups similar embeddings into **k=5** clusters, matching the number of semantic categories.

PCA Reduction

Projects 384-dimensional embeddings into **2D** for visual inspection of semantic structure.

Results

1.0000

Adjusted Rand Index

Perfect cluster-label agreement

1.0000

NMI Score

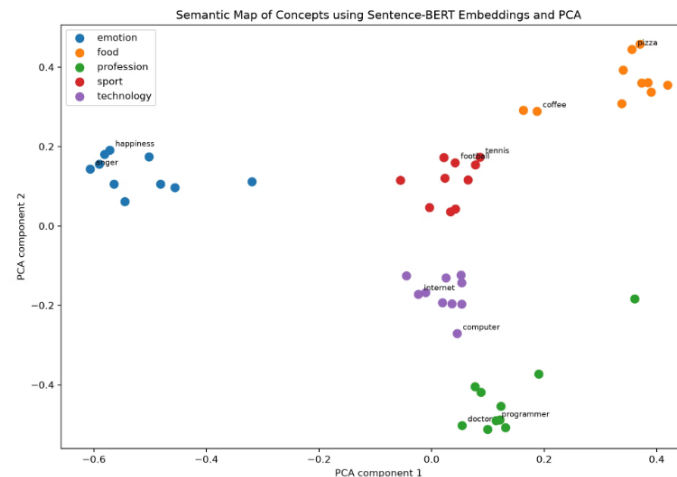
Normalized Mutual Information

0.2927

Silhouette Score

Moderate inter-cluster separation

K-Means successfully **recovered all five semantic categories** with perfect alignment to reference labels. The silhouette score reflects moderate separation — expected given the compact, well-defined nature of the dataset.



Discussion

Key Findings

- Sentence-BERT captured **contextual semantic relationships** effectively
- Concepts from the same category clustered together with high fidelity
- ARI and NMI of **1.0** indicate perfect agreement with ground-truth labels
- Silhouette Score (0.29) suggests moderate but acceptable cluster separation

Project Strengths

**Contextual
Embeddings**

**Semantic
Understanding**

Modular Pipeline

Reproducible

Conclusion

This project demonstrates that **Sentence-BERT embeddings** can successfully generate semantic representations capable of organizing textual concepts into meaningful, interpretable clusters – validating the core research question.

Embedding Extraction

Sentence-BERT transforms text into dense semantic vectors

Clustering & Visualization

K-Means and PCA reveal semantic structure in 2D space

Full Pipeline

End-to-end reproducible workflow from raw text to evaluation

Thank You

